

Ruijie Xu

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SUMMARY

Master's student in Data Science with experience applying data-driven approaches to complex economic and infrastructure problems. Skilled in structuring ambiguous problems, integrating multi-source data, and translating analytical insights into actionable decisions.

EDUCATION

Georgetown University | 2024 – 2026 | Master of Science in Data Science & Analytics | GPA: 3.8 / 4.0

Soochow University | 2018 – 2023 | Bachelor of Architecture | GPA: 3.5 / 4.0

Coursework: Statistical Learning | Data Visualization & Storytelling | Generative AI Applications

EXPERIENCE

Design Assistant | Tus-Design Group Co., Ltd. | Suzhou City | Jun – Aug 2021

- Synthesized land use, policy, and stakeholder inputs into structured design proposals for an urban innovation district
- Structured complex design decisions into client-facing presentations to support stakeholder review and approval
- Benchmarked design typologies and cost-feasibility trade-offs to strengthen the firm's competitive bid positioning

Teaching Assistant | Statistical Learning | Georgetown University | Jan – May 2026

- Identified recurring failure patterns in student model implementations and provided structured feedback to improve analytical reasoning
- Synthesized common errors across submissions into targeted guidance, helping students improve model interpretation and evaluation quality
- Coordinated with faculty to align evaluation standards and ensure consistency across a large cohort

Research Assistant | Remote | Prof. Ye | Soochow University | Jan 2026 – Present

- Designed a multi-channel analytical framework decomposing time-series behavioral data into spatial and activity dimensions, enabling systematic pattern detection across two provincial datasets
- Structured the research methodology by selecting and sequencing unsupervised ML techniques to surface distinct public property usage clusters
- Collaborated across data science and architectural research domains to translate behavioral sequence findings into built-environment insights

SELECTED PROJECTS

U.S. Manufacturing Landscape Analysis

- Aggregated publicly available datasets across U.S. manufacturing sectors from government sources, identifying sector-level patterns through data visualization and narrative analysis
- Designed and built an automated ingestion pipeline that periodically extracts updated government data, eliminating manual refresh and ensuring dashboards always reflect the latest
- Developed a fixed-template dashboard presenting key manufacturing metrics in a consistent visual style, enabling repeatable, client-ready storytelling on sector performance over time

AI Policy Memo Agent with Automated Fact Verification

- Identified a critical gap in LLM-assisted policy research that agents micro-decide what evidence to include with no infrastructure to catch hallucinated statistics or false causal claims before they propagate into institutional decisions
- Designed a four-phase human-in-the-loop pipeline with a structured 6-type claim taxonomy, routing each claim to the appropriate validation tier based on its epistemic structure
- Achieved 96% accuracy on the holdout benchmark, resolving 90–96% of claims automatically at 23% of the cost of a single-model baseline

Data Center Site Selection Analysis

- Synthesized 4,428+ data center locations and 10+ county-level official datasets into a unified analytical framework, surfacing actionable siting signals across geographic, regulatory, and infrastructure dimensions for all states
- Developed an LLM-based policy classifier with human-in-the-loop validation to extract and label county regulatory environments, identifying 80+ high-potential candidate counties from scraped policy text
- Engineered 38 county-level features keyed by FIPS codes to support predictive modeling of site suitability, enabling a structured, data-driven approach to infrastructure location decisions

SKILLS

Programming: Python | SQL | R

Analytics: Exploratory Data Analysis | Statistical Modeling | Data Visualization

Machine Learning: Supervised Models | Feature Engineering | Model Evaluation

Generative AI: LLM Integration | Prompt Engineering

Tools: Jupyter | Git | Docker

LANGUAGES

English (Proficient) | Mandarin (Native)